

SAKSHI SANJAY BHARAMBE

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EDUCATION

University of Southern California

Los Angeles, CA

Master of Science in Computer Science

August 2024 – May 2026

- Coursework: Analysis of Algorithms, Web Technologies, Multimodal Learning, Applied NLP, Information Retrieval and Web Search Engines, Research Methods and Analysis for User Studies, Deep Learning, Database Systems

Terna Engineering College, University of Mumbai

Mumbai, India

Bachelor of Engineering in Computer Engineering

August 2019 – June 2023

- Coursework: Artificial Intelligence, Machine Learning, Applied Data Science, Big Data Analytics, Software Engineering, Data Structures, Computer Networks

SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, C, HTML, CSS, PHP, Swift

Frameworks: React, Node.js, Express.js, Selenium, Flask, FastAPI, Bootstrap, Streamlit

Libraries: TensorFlow, PyTorch, Scikit-learn, LangChain, Hugging Face Transformers, OpenCV, MediaPipe, Pandas, NumPy

Databases: PostgreSQL, pgvector, MySQL, MongoDB, Redis

Tools and Platforms: Git, GitHub, Docker, AWS, Azure AD, VS Code, Jupyter Notebook, PyCharm, Google Colab, Ollama

Engineering Concepts: RAG, AI Agents, Web Scraping, Automation, Multithreading, REST APIs, OOP, Data Pipelines

EXPERIENCE

USC Transportation

Los Angeles, CA

Software Developer

September 2024 – May 2025

- Automated data-intensive workflows for internal operations by developing Python scripts with Selenium and Pandas, reducing manual data entry tasks and boosting processing efficiency by 75%
- Designed a scalable web scraping and auto-fill pipeline with multithreaded execution, implementing error handling and logging to enhance data reliability and increase reservation throughput fivefold
- Implemented Microsoft Outlook notifications into enterprise systems via Azure AD and Python-based APIs, facilitating real-time alerts and reduce communication delays by 40% across operational teams

Leap Analytics

Los Angeles, CA

Software Engineer Intern

May 2025 – August 2025

- Improved Leap's existing website by adding new pages, refining UI components, and enhancing responsiveness with HTML, CSS, and JavaScript, increasing client engagement by 20%
- Constructed automated web scraping pipelines using Python and Selenium to extract structured datasets from multiple sources, decreasing data collection time by 80% and accelerating market research
- Performed data cleaning, exploratory analysis, and visualization with Pandas, NumPy, and Matplotlib, improving reporting accuracy by 25% and enabling more reliable strategy decisions

Blend Vidya

Mumbai, India

Artificial Intelligence Intern

January 2023 – June 2023

- Analyzed 10,000+ student activity records to derive engagement patterns by applying data cleaning and visualization using Pandas, NumPy, and Matplotlib, improving reporting accuracy by about 30%
- Integrated AI-driven insights by developing intelligent data visualization tools and feedback analytics, enhancing mentor-student interactions and increasing learner satisfaction by approximately 20%
- Led a team of 4 interns to create a predictive analytics module for course recommendation by leveraging clustering and classification models, boosting student engagement by 25% based on personalized suggestions

PROJECTS

Research Paper Assistant | *FastAPI, RAG, Llama 3*

- Engineered a fully local RAG application using FastAPI, PostgreSQL with pgvector, MiniLM embeddings, and Llama 3 via Ollama, enabling grounded question answering over research papers with clickable page citations and complete data privacy
- Built a 15-question evaluation pipeline that exposed 5 retrieval failures, then implemented first-page routing and per-document retrieval using SQL window functions, resolving title, author, and method question failures

LoomLog | *LangChain, React, MongoDB*

- Developed a full-stack life management platform with a React frontend, Express REST API, MongoDB, and JWT and bcrypt authentication, unifying tasks, calendar, and finances with secure per-user data isolation at zero cost
- Implemented a LangChain tool-calling agent with 3 scoped tools that lets users query and update tasks, events, and finances in natural language, computing balances in code instead of the LLM to prevent arithmetic errors

Multimodal Emotion Recognition System | *BERT, HuBERT, Streamlit*

- Designed a multimodal emotion recognition pipeline using BERT, HuBERT, and EfficientNet-B3 to process text, audio, and video inputs, achieving 60% accuracy through weighted late fusion
- Developed a Streamlit-based inference application that automated preprocessing, feature extraction, and emotion prediction, enabling end-to-end emotion recognition from uploaded videos through a unified inference pipeline

QueryBot | *BERT, T5, SQL*

- Built a scalable preprocessing pipeline for the Spider dataset by standardizing data cleaning, schema linking, and JSON/CSV serialization, enabling efficient training of transformer-based Text-to-SQL models
- Optimized schema-aware preprocessing workflows and interactive data visualizations, reducing query parsing errors by 30% and improving SQL generation across diverse database schemas